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Which AI Tool for Which Job

A map of AI tools for students of mathematics, grouped by what you need to do.

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Pick the tool for the job, not the other way round. A chat assistant is good at explaining and bad at arithmetic; a computer algebra system is the opposite. Facts below come from each tool's official pages and were last checked in November 2026. These tools change quickly, so check the page before you rely on a feature.

Explaining and tutoring: general assistants

ChatGPT chatgpt.com	General assistant from OpenAI. Study mode asks questions and gives hints instead of the final answer.
Gemini gemini.google.com	General assistant from Google. Guided Learning walks you through a problem step by step.
Grok grok.com	General assistant from xAI (now SpaceXSI), with real-time answers from the web and X.
Claude claude.com	General assistant from Anthropic.

Watch out: They write wrong answers as fluently as right ones. Age rules differ: ChatGPT and Grok are 13+ with a parent's permission under 18, Gemini needs a Google Account you may manage yourself (14+ in Italy), and Claude is 18+.

Studying from your own material

Gemini Notebook (formerly NotebookLM) notebook.google	Answers questions about the notes, slides, and PDFs you upload, cites the passages it used, and makes study guides, flashcards, and quizzes.

Watch out: Answers come only from what you upload, so a gap in your notes is a gap in the answer. Open the cited passage before you trust a summary.

Computing and checking

Wolfram|Alpha wolframalpha.com	Solves equations, limits, and integrals from plain-language input. Step-by-step solutions need Pro.
GeoGebra CAS Calculator geogebra.org/cas	Free computer algebra in the browser: solve, factor, differentiate.
Desmos Graphing Calculator desmos.com/calculator	Plot a function to check zeros, signs, and asymptotes.

Watch out: These compute exactly, but only what you typed. Check that the input is the problem you meant.

University work: LaTeX, code, papers, proofs

Overleaf AI features docs.overleaf.com	Error Assist explains LaTeX errors and suggests fixes. Free within daily limits.
Gemini in Colab developers.google.com/colab	Python notebooks where Gemini generates, explains, and debugs code. Learn Mode guides you instead of writing the code.
Semantic Scholar semanticscholar.org	Search research papers. TLDRs give a one-sentence AI summary of each result.
Lean and Mathlib lean-lang.org	Advanced: a proof assistant that checks every step of a proof, and its library of formalized mathematics (mathlib.org).

Watch out: Read every line of generated LaTeX or code before you use it. A one-sentence summary is not the paper.

Coding agents

Grok Build
x.ai/build
xAI's coding agent for the terminal: plans, edits files, runs builds and tests. Free to try.

Claude Code
claude.com/product/claude-code
Anthropic's coding agent for the terminal, editors, and the web. Asks before it changes files or runs commands. Paid Claude plans.

Codex
chatgpt.com/codex
OpenAI's coding agent in ChatGPT, your editor, and the terminal, with cloud environments for each repository. Included in paid ChatGPT plans.

Google Antigravity
antigravity.google
Google's agentic development platform: an app, a CLI, an IDE, and editor extensions where agents write, run, and test code.

Watch out: They edit and run code on your machine or in your repository. Review every change, use version control, and do not hand them graded coursework.

Agents and personal assistants

Agents do not just answer: they carry out multi-step tasks such as browsing, running code, organising files, and scheduling.

Grok Bot
x.ai/bot
Agents with their own cloud computer that use apps and websites, write and run code, and work on a schedule.

OpenAI Dots
chatgpt.com/features/dots
Always-on ChatGPT agents with their own cloud computer that keep working on your projects between conversations. Paid plans, 18+, not yet available in the EU.

Meta Muse
about.fb.com
Meta's personal agent on its own cloud computer: it browses, fills in forms, and books things, and asks before it sends an email or buys. Rolling out in the US, free with a usage limit.

Claude Duet
claude.com/product/claude-duet
Anthropic's proactive agent on its own cloud computer that works in your connected apps and asks before it sends, pays, or deletes. US research preview for paid plans, by waitlist.

Watch out: Agents act on your accounts and can be confidently wrong. Review what they do, keep personal data and school accounts out, and never let them submit graded work.

Quick map

I want to …	Reach for
understand an idea or get a hint	a general assistant in study or tutor mode
revise from my own notes	Gemini Notebook
check an answer or a calculation	Wolfram|Alpha, GeoGebra CAS, Desmos
fix a LaTeX error	Overleaf Error Assist
understand or debug Python	Gemini in Colab
work on a coding project	a coding agent, inside version control
find papers on a topic	Semantic Scholar, then read the papers
be certain a proof is correct	Lean, or a careful human reader
hand off a multi-step chore	an agent, and check everything it did